Controlled comparison of six resampling conditions on 30 binary datasets

Supervisor briefing — UGO-SMOTE, BFCH-SMOTE, EC-EBO, SSHR, SMOTE and no resampling

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Purpose and main finding

This study compares four implemented paper-inspired methods—UGO-SMOTE, BFCH-SMOTE, EC-EBO and SSHR—against standard SMOTE and a no-resampling control on the **same 30 binary dataset variants**. The results show metric and classifier tradeoffs, not a universal winner. Standard SMOTE has the highest variant-average LogReg F1 (0.495) and G-mean (0.800); UGO-SMOTE has the highest variant-average GNB F1 (0.505) and MCC (0.486). The no-resampling control has the lowest average false-positive rate for both classifiers. In the SSHR ablation, supplying unlabelled training examples slightly raises G-mean but lowers F1 and raises FPR on average.

These are **descriptive results from our controlled implementation**, not a replication of the four papers' published scores. No new method or novelty claim is established here.

Experimental setup

- **Inputs:** 30 numeric, no-missing binary variants from 10 source families; 9 Ecoli and 10 Yeast variants share related original data. Satellite class 4 versus rest and Abalone rings ≥18 versus rest are explicitly accepted approximations. Exact author preprocessing and folds are unverified.
- **Evaluation:** Five repeats of five stratified, coordinate-group-disjoint outer folds, seed 42. Within each outer training fold, approximately half of each class's rows remain labelled (`L`); the others form a training-only unlabelled pool (`U`). No outer-test row is used to fit preprocessing, resampling or a classifier.
- **Common comparison:** NORES, SMOTE, UGO-SMOTE, BFCH-SMOTE, EC-EBO and SSHR use the same `L`, held-out fold and final classifier. Only SSHR may use `U`. Classifiers are logistic regression (`LogReg`) and Gaussian naive Bayes (`GNB`); their scores are reported separately.
- **SSHR ablation:** SSHR and NORES are also run without `U` on the same folds and `L`. The saved NORES outputs matched exactly for all 60 dataset/classifier conditions, supporting a controlled comparison.
- **Measures:** F1, G-mean, MCC and probability ROC-AUC are higher-is-better; FPR is lower-is-better. Each cell below is the unweighted average of 30 dataset-level means. Each dataset's six main methods use its same successful fold subset.

Source-family counts are Yeast 10, Ecoli 9, Glass 3, Wine Quality 2, and one each of Banana, Satellite, Abalone, New Thyroid, Page Blocks and Shuttle.

Execution and coverage audit

The upload has 30 report folders, 1,800 canonical score rows (30 datasets × 2 classifiers × 6 methods × 5 metrics), and 300 SSHR ablation rows. The ZIP passed CRC and every included file matched its recorded SHA-256. The selection and source-registry hashes match the Windows checkout's CRLF line endings. The aggregate CSV matches the canonical rows in all 30 per-dataset score files exactly. This audit checks the **aggregate package**; individual saved job records and predictions were not uploaded.

Run	Planned jobs	Successful	Failed
Six-method main comparison	9,000	8,994	6
SSHR without-U ablation	3,000	3,000	0

All six failures are standard SMOTE on `winequality_red_8_vs_6`: the same three folds fail under both classifiers because five labelled minority fitting rows are insufficient for its fixed five-neighbour setting (`n_neighbors = 6` requires at least six rows). The study did not change the sampler or redraw folds after seeing this. The six-way table uses **25 shared folds for 29 variants** and only **22 shared folds for this wine variant**. Its SSHR with/without-U ablation uses all 25 paired folds, so those two summaries have different fold scopes. The rarest test fold for this variant has one minority row, adding instability.

Main comparison

Classifier	Method	F1 ↑	G-mean ↑	MCC ↑	ROC-AUC ↑	FPR ↓
LogReg	NORES	0.426	0.487	0.431	0.866	0.008
LogReg	SMOTE	0.495	0.800	0.479	0.868	0.137
LogReg	UGO-SMOTE	0.475	0.582	0.462	0.862	0.047
LogReg	BFCH-SMOTE	0.489	0.778	0.466	0.858	0.126
LogReg	EC-EBO	0.492	0.794	0.473	0.868	0.125
LogReg	SSHR	0.421	0.689	0.406	0.856	0.119
GNB	NORES	0.487	0.653	0.471	0.846	0.061
GNB	SMOTE	0.482	0.771	0.465	0.838	0.153
GNB	UGO-SMOTE	0.505	0.685	0.486	0.847	0.074
GNB	BFCH-SMOTE	0.442	0.712	0.418	0.810	0.125
GNB	EC-EBO	0.439	0.762	0.421	0.832	0.167
GNB	SSHR	0.424	0.668	0.410	0.839	0.112

Reading the tradeoff. Under LogReg, SMOTE gains 0.069 F1 and 0.314 G-mean over NORES on average, but increases FPR by 0.128. UGO-SMOTE gains 0.048 F1 with a smaller 0.039 FPR increase; its G-mean gain is only 0.096. BFCH-SMOTE and EC-EBO have average LogReg F1 of 0.489 and 0.492, close to SMOTE's 0.495, with slightly lower FPR but no overall F1 advantage. Under GNB, UGO-SMOTE improves average F1 over NORES by 0.018 while SMOTE is lower by 0.005. EC-EBO and BFCH-SMOTE have lower average GNB F1 than both UGO-SMOTE and NORES.

These means conceal heterogeneity: UGO-SMOTE has the highest LogReg F1 on 13 of 30 variants even though SMOTE has the higher variant-average F1. A method's performance depends on the classifier and dataset. FPR should be read alongside recall-sensitive metrics; a very conservative model can achieve low FPR while missing minority examples.

Questions addressed by the methods

Boundary sampling and overlap (BFCH-SMOTE). BFCH-SMOTE applies an observed-majority exclusion rule during its geometric generation. That does not guarantee low overlap on unseen data. The study's *training-only proxy* asks whether a generated minority point is nearer an observed majority row than an observed minority row. BFCH-SMOTE has a higher proxy rate than SMOTE on **26 of 30** variants, a lower rate on 3, and a tie on 1; its variant-average proxy rate is **0.141 versus 0.102** for SMOTE. These rates do not measure the true decision boundary or true class overlap. The current results therefore do **not** demonstrate the desired reduction in overlap.

Intra-class imbalance (EC-EBO). EC-EBO's entropy and clustering mechanism is implemented, but this comparison does not separately label minority subclusters or measure improvement by subcluster. Its mean LogReg G-mean is 0.794 versus SMOTE's 0.800; GNB G-mean is 0.762 versus 0.771. This study is insufficient to claim that it resolves intra-class imbalance, although dataset-specific improvements may warrant inspection.

Unlabelled data (SSHR). SSHR uses `U` to select and pseudo-label existing training observations and may generate new minority vectors through its SMOTE fallback. The ablation estimates what supplying `U` changed under the fixed protocol:

Classifier	Metric	SSHR with U	SSHR without U	With − without
LogReg	F1	0.421	0.438	-0.016
LogReg	G-mean	0.689	0.676	+0.013
LogReg	MCC	0.406	0.422	-0.016
LogReg	ROC-AUC	0.856	0.861	-0.005
LogReg	FPR	0.119	0.098	+0.021
GNB	F1	0.425	0.451	-0.026
GNB	G-mean	0.669	0.657	+0.012
GNB	MCC	0.411	0.437	-0.026
GNB	ROC-AUC	0.840	0.839	+0.001
GNB	FPR	0.112	0.089	+0.024

Supplying `U` lowers variant-average F1 by 0.016 (LogReg) and 0.026 (GNB), while G-mean rises by about 0.013 and 0.012. FPR rises by about 0.022–0.024. F1 declines on 22/30 LogReg variants and 23/30 GNB variants, with one tie in each. Thus the present evidence does not support a general F1 benefit from `U`; it shows a recall/false-positive tradeoff. These ablation means use all 25 paired folds, including the three folds excluded from the six-method wine comparison.

Interpretation limits and next discussion

The 30 entries are not 30 independent original datasets: Ecoli and Yeast alone contribute 19 entries. The averages are descriptive variant averages, not confidence intervals or independent-sample statistical tests. Repeated folds share observations, and the study uses only two classifiers, one labelled fraction, fixed method defaults and one seed. Paper-specific data preparation, hyperparameters, classifier sets and SSHR's original unlabelled-pool protocol are not established here. UGO-SMOTE remains a provisional, source-audited implementation; BFCH-SMOTE includes declared geometric repairs; EC-EBO is an audited interpretation; SSHR also contains explicit implementation choices. Consequently, differences from original paper tables are expected.

For a subsequent research method, define the primary target **before** tuning—for example, minority F1 subject to a maximum FPR—and evaluate boundary quality with a prespecified held-out or geometric overlap measure. The current BFCH proxy and SSHR ablation identify concrete limitations to discuss with the supervisor; they do not by themselves establish a new algorithm.

Files and provenance

- Three-page figure pack and figure-reading guide: visual summaries of F1/FPR tradeoffs, per-variant consistency and the SSHR ablation.
- 30-dataset comparison table: 300 Excel-friendly rows, with all five metrics side by side for six methods, separately by classifier and dataset.
- Detailed aggregate scores: 1,800 rows with mean, sample SD, shared-fold count and paired difference from NORES.
- Method coverage: planned, successful and failed fold events for every dataset, classifier and method.
- SSHR unlabelled-pool ablation: paired with/without-U effects for each dataset, classifier and metric.
- Study protocol and fixed selection.
- Uploaded review ZIP SHA-256: `05461d8e67ff55741982a0b8115fba80fcf6af0c7ff1ae8804e70699cca2b4be`. It is retained by the student as the original aggregate report bundle. Raw datasets and individual job outputs remain local.

Paper sources: Sağlam & Cengiz, UGO, *Knowledge-Based Systems* (2026), DOI 10.1016/j.knosys.2026.115886; Wen et al., BFCH-SMOTE, *Neurocomputing* (2026), DOI 10.1016/j.neucom.2026.134595; Tao et al., EC-EBO, *Knowledge-Based Systems* (2026), DOI 10.1016/j.knosys.2026.115544; Jiang et al., SSHR, *Expert Systems with Applications* (2023), DOI 10.1016/j.eswa.2023.119733. This report cites the studied methods; it does not assert exact author-code equivalence.